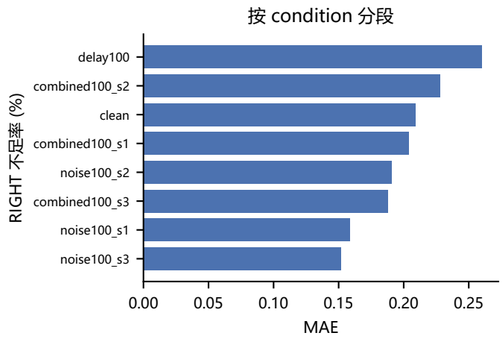
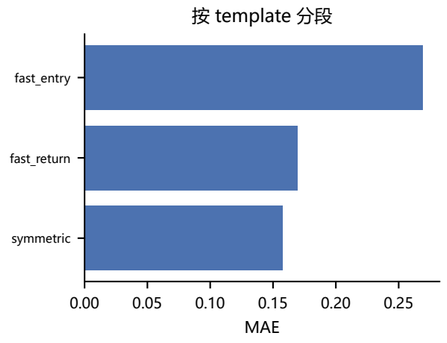
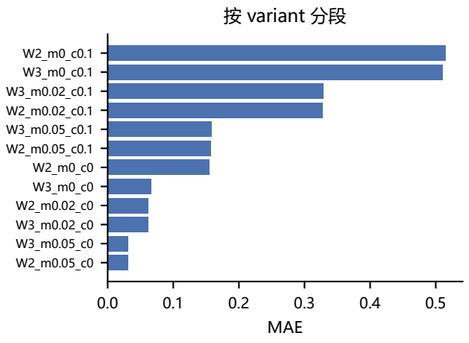
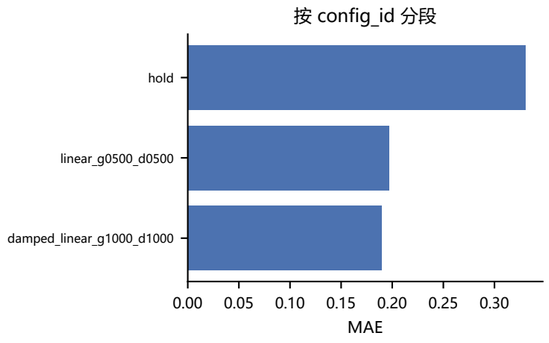
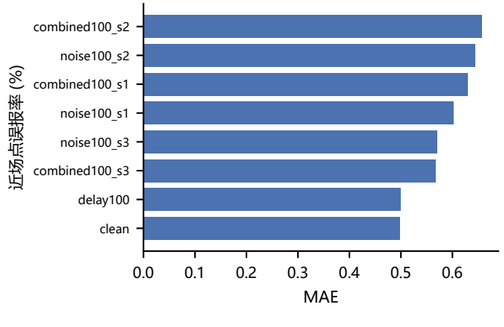
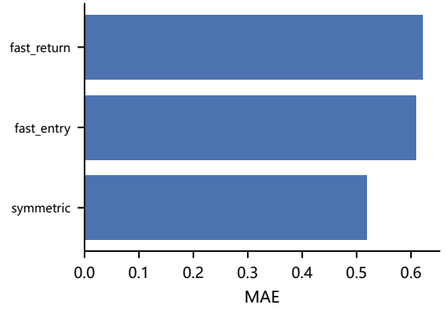


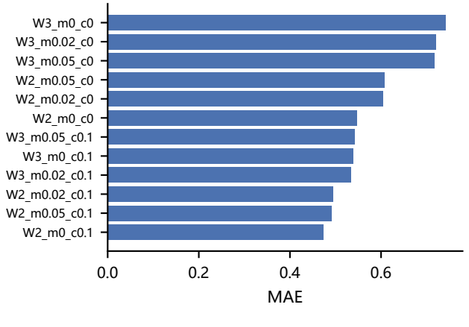


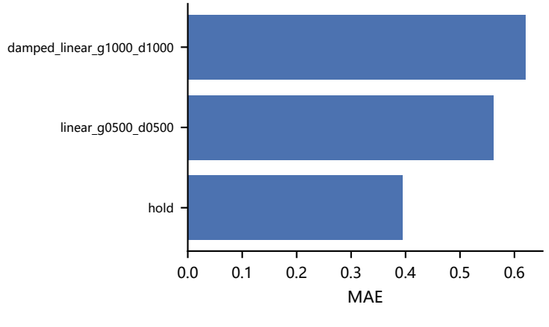


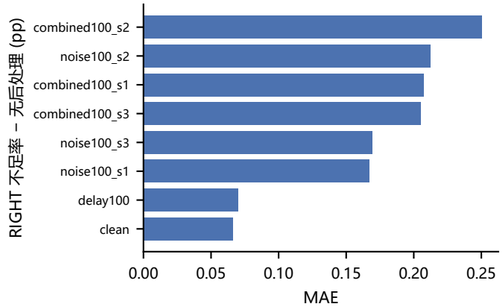


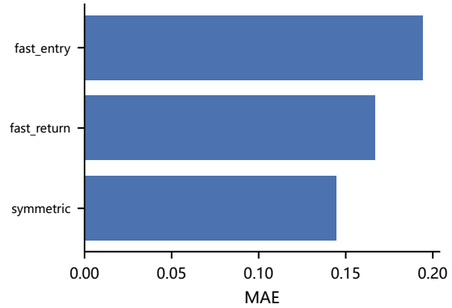


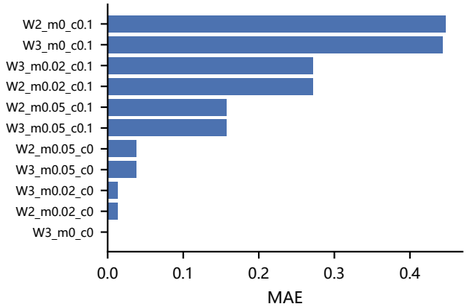


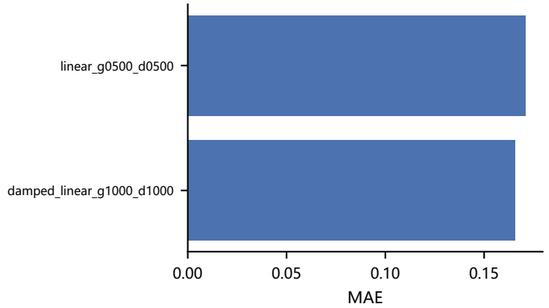


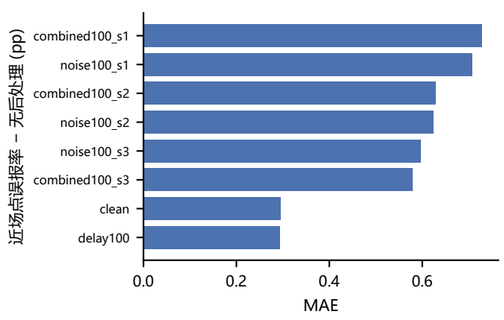


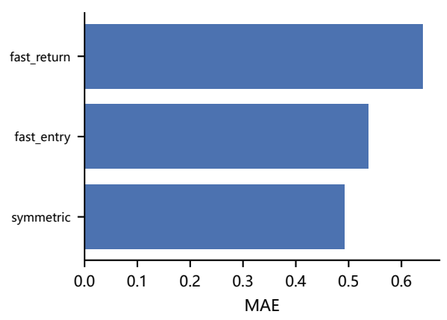


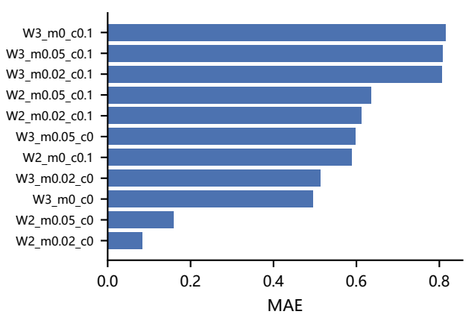


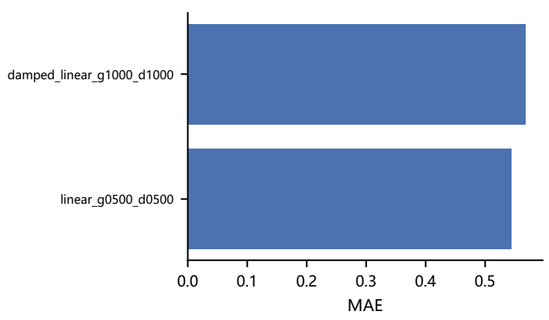












中期答辩临时训练：基于接触标签纠错前评分，非冻结科研资产。 数据 S3（修正评分口径），训练 27 条开发轨迹，外部检验 12 条选择轨迹，转角峰值均为 10°。 主方案折外误差按分段统计；最难预测的前 10 个轨迹×条件见 results/segment_errors.csv。